

Sample Admissions Test

Paper 1

Sample Set 3 (2026)

Time allowed: 1 hour 15 minutes

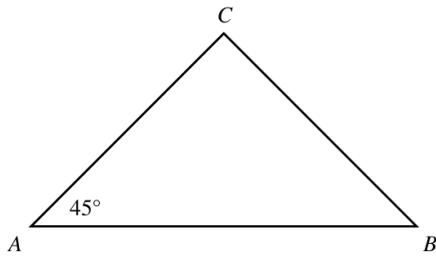
There are 8 questions. Answer all of them.

1. Find the complete set of values of x for which

$$\frac{x^2 - 4}{x + 1} \leq 0$$

- A $x \leq -2$
- B $-2 \leq x < -1$ or $x \geq 2$
- C $x \leq -2$ or $-1 < x \leq 2$
- D $-1 < x \leq 2$
- E $x \geq 2$

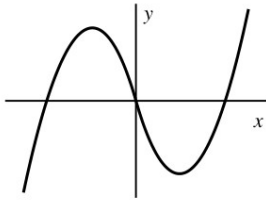
2. The diagram shows triangle ABC with $AB = 9/\sqrt{2}$ and angle $BAC = 45^\circ$.



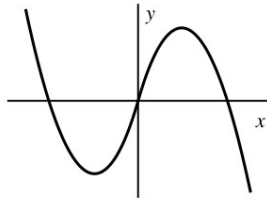
What is the area of triangle ABC ?

- A $\frac{81}{4}$
- B $\frac{81}{2}$
- C $\frac{81\sqrt{2}}{4}$
- D 81

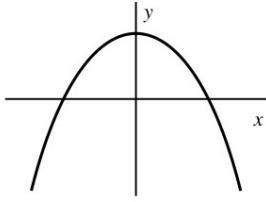
3. Which one of the following could be the graph of $y = x^3 - x$?



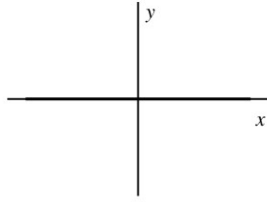
A



B



C



D

4. Evaluate

$$\sum_{k=1}^{10} \log_2 \left(\frac{k+1}{k} \right)$$

- A $\log_2 11$
- B $\log_2 10$
- C $1 + \log_2 11$
- D 11
- E $\log_{10} 2$

5. Consider the following statements about the function $f(x) = x - 1 \vee -2$:

- I $f(-1) = 0$
- II $f(x) \geq -2$ for all real x
- III the equation $f(x) = 1$ has exactly two solutions

Which of these statements is/are true?

- A none of them
- B I only
- C II only
- D I and II only
- E I, II and III

6. Given that $\int_0^a (3x^2 - 2x) dx = 4$ where $a > 0$, find the value of a .

A 1

B 2

C $\frac{1+\sqrt{17}}{2}$

D $\sqrt{3}$

E 4

7. The sequence u_n is defined by $u_1 = 2$ and $u_{n+1} = \frac{u_n}{1+u_n}$ for $n \geq 1$.

What is the value of u_{10} ?

A $\frac{2}{19}$

B $\frac{1}{10}$

C $\frac{2}{21}$

D 2^{-10}

E $\frac{1}{2^9}$

8. The table shows some values of the function f .

x	0	1	2	3
f(x)	1	3	5	7

Which one of the following could be $f(x)$?

A $2x+1$

B x^2+1

C 3^x

D $\sqrt{x+1}$

E $\frac{6}{x+1}$

END OF TEST